

AIM:

Upload and analyse the given dataset and perform data preprocessing techniques and visualisation to find the outliers.

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
data = {
    "Age": [20, 21, 22, 23, 21, 24, 22, 25, 23, 22, 21, 100],
    "Salary": [25000, 28000, 30000, 32000, 29000, 31000,
               30500, 33000, 31500, 29500, 28500, 150000]
}
df = pd.DataFrame(data)
for column in df.select_dtypes(include="number").columns:
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    outliers = df[
        (df[column] < lower_bound) |
        (df[column] > upper_bound)
    ]
    print("\nColumn:", column)
    print("Q1:", Q1)
    print("Q3:", Q3)
    print("IQR:", IQR)
    print("Lower Bound:", lower_bound)
    print("Upper Bound:", upper_bound)
    print("Outliers:")
    print(outliers[[column]])
plt.figure(figsize=(8, 5))
df.boxplot(column=["Age", "Salary"])
plt.title("Detection of Outliers Using Boxplot")
plt.ylabel("Values")
plt.show()
```

OUTPUT:

Column: Age

Q1: 21.0

Q3: 23.25

IQR: 2.25

Lower Bound: 17.625

Upper Bound: 26.625

Outliers:

Age
11 100

Column: Salary

Q1: 28875.0

Q3: 31625.0

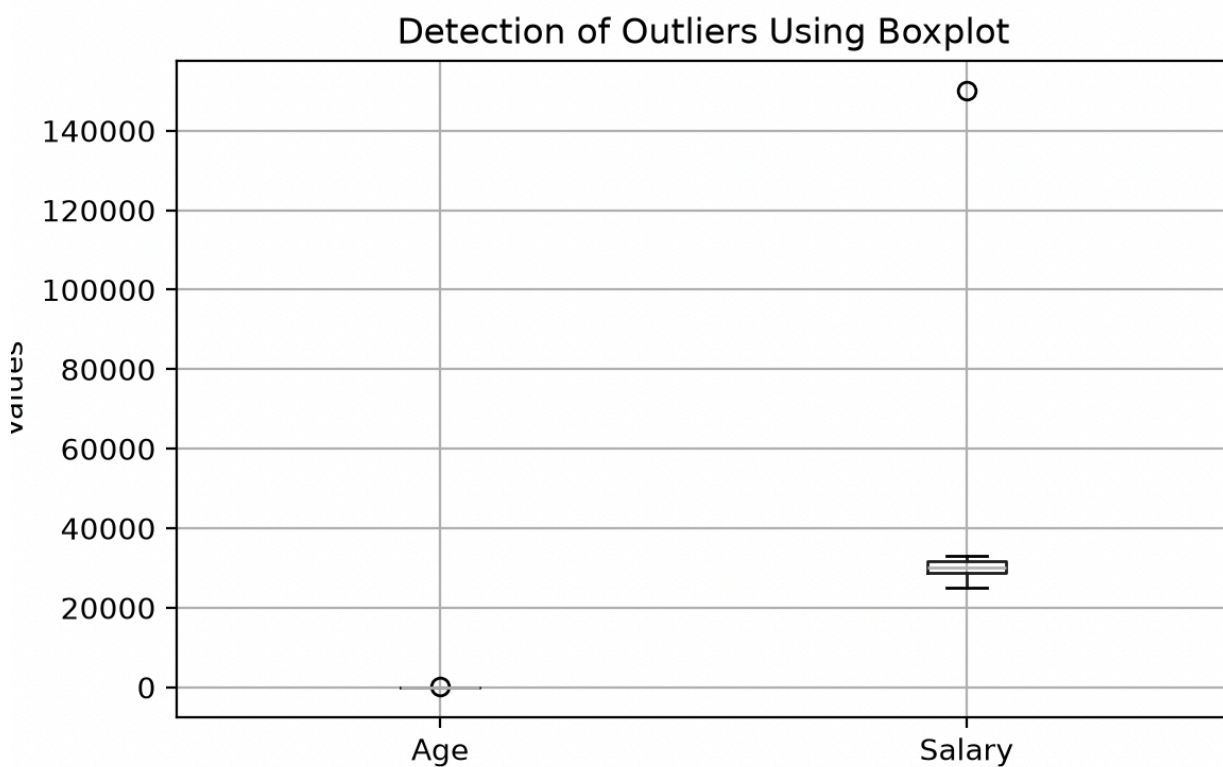
IQR: 2750.0

Lower Bound: 24750.0

Upper Bound: 35750.0

Outliers:

Salary
11 150000



RESULT:

The given dataset is analysed and preprocessed successfully.